

DATABASE MANAGEMENT SYSTEMS LAB

II B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS07	PCC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Apply the basic concepts of Database Systems and Applications.
2. Use the basics of SQL and construct queries using SQL in database creation and interaction
3. Design a commercial relational database system (Oracle, MySQL) by writing SQL using the system.
4. Analyze and Select storage and recovery techniques of database system.

COURSE OUTCOMES:

Upon successful completion of this course, student will be able to:

1. Apply the basic concepts of Database Systems and Applications.
2. Develop an ER model for a given database.
3. Use the basics of SQL and construct queries using SQL in database creation and interaction.
4. Design a commercial relational database system (Oracle, MySQL) by writing SQL using the system.

LIST OF EXPERIMENTS

WEEK-1 INTRODUCTION TO E R MODEL

Student should select a case study from given ideas and formulate the problem statement.

WEEK-2 CONCEPTUAL DESIGNING USING ER DIAGRAMS

Conceptual Designing using ER Diagrams (Identifying entities, attributes, keys and relationships between entities, cardinalities, generalization, specialization etc.) Note: Student is required to submit a document by drawing ER Diagram to the Lab teacher.

WEEK-3 CONVERTING ER MODEL TO RELATIONAL MODEL

Converting ER Model to Relational Model (Represent entities and relationships in Tabular form, Represent attributes as columns, identifying keys) Note: Student is required to submit a document showing the database tables created from ER Model.

WEEK-4 NORMALIZATION

Normalization -To remove the redundancies and anomalies in the above relational tables, Normalize up to Third Normal Form

WEEK-5 CREATION OF TABLES USING SQL

Creation of Tables using SQL- Overview of using SQL tool, Data types in SQL, Creating Tables (along with Primary and Foreign keys), Altering Tables and Dropping Tables

WEEK-6 DML COMMANDS

Practicing DML commands- Insert, Select, Update, Delete.	
WEEK-7	QUERIES
Practicing Queries using ANY, ALL, IN, EXISTS, NOT EXISTS, UNION, INTERSECT, CONSTRAINTS etc.	
WEEK-8	SUB QUERIES
Practicing Sub queries (Nested, Correlated) and Joins (Inner, Outer and Equi)..	
WEEK-9	AGGREGATE FUNCTIONS
Practice Queries using COUNT, SUM, AVG, MAX, MIN, GROUP BY, HAVING, VIEWS Creation and Dropping.	
WEEK-10	TRIGGERS
Practicing on Triggers - creation of trigger, Insertion using trigger, Deletion using trigger, Updating using trigger.	
WEEK-11	PROCEDURES
Procedures- Creation of Stored Procedures, Execution of Procedure, and Modification of Procedure	
WEEK-12	CURSORS
. Cursors- Declaring Cursor, Opening Cursor, Fetching the data, closing the cursor.	
REFERENCE BOOKS:	
1. Database System Concepts, by Silberschatz, Sudarshan, and Korth, 6th edition. 2. Database management System by RaghuRamaKrishna, 3rd edition.	
WEB REFERENCES:	
1. http://www.learnadb.com/databases/how-to-convert-er-diagram-to-relational-database 2. https://www.w3schools.com/sql/sql_create_table.asp 3. http://www.edugrabs.com/conversion-of-er-model-to-relational-model/?upm_export=print 4. http://ssyu.im.ncnu.edu.tw/course/CSDB/chap14.pdf 5. http://web.cs.ucdavis.edu/~green/courses/ecs165a-w11/8-query.pdf	